

Supplementary Material A: Computational Supplement for “Spatio-Temporal Models under Spatial Regimes: the Spatially-Clustered STEM Family and its Implementation in the **Stem** R Package”

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This document collects the computational derivations and results accompanying the main paper; it is also the technical documentation of the estimation in the **Stem** package. Section [S1](#) states how the package estimates the model: the Kalman filter and smoother, the EM algorithm for STEM and within SC-STEM, where it is slow, the ECME and SQUAREM algorithms that accelerate it, the stopping rule and its relation to that of D-STEM, the limits within which the range is estimated, and the numerical experiments behind these choices. Section [S2](#) derives the completed sufficient statistics that the EM algorithm requires when the response is incompletely observed, which the manuscript states without proof. Section [S3](#) derives the behaviour of the information criteria as the dimensions grow. Sections [S4](#) and [S5](#) document the ridge as the package implements it: the exact scale of λ , the switches that reduce the model to penalized linear regression, the relation between one λ and the regimes, and the order in which the hyperparameters have to be chosen. Section [S6](#) states where a fit starts — from the departures of the locations from the pooled model when it is unpenalized, from the unpenalized solution when it is penalized — and where the scale of the penalty is measured. Section [S7](#) points to Supplementary Material B, which collects all the material of the simulation study. Section [S8](#) documents the reproduction code. All figures and tables are generated by the analysis scripts.

S1 Estimation: the Kalman filter, the EM algorithm and its accelerations

This section states the estimation of STEM and SC-STEM as the **Stem** package implements it: the state-space form of the model (Section [S1.1](#)), the Kalman filter and smoother on which every algorithm rests (Section [S1.2](#)), the EM algorithm for STEM and its use within SC-STEM (Sections [S1.3](#) and [S1.4](#)), where it is slow (Section [S1.5](#)), the two accelerations the package offers, ECME and SQUAREM (Sections [S1.6](#) and [S1.7](#)), their implementation (Section [S1.8](#)), the stopping rule (Section [S1.9](#)), the treatment of the range at the boundary of what the data identify (Section [S1.10](#)), a summary of the four algorithms (Section [S1.12](#)) and the numerical experiments that support the choices (Section [S1.13](#)).

S1.1 The model in state-space form

Within a regime, or in the pooled fit, STEM is the state-space model

$$z_t = X_t \beta + A y_t + e_t, \quad e_t \sim N(0, \Sigma_e), \quad \Sigma_e = \sigma_\omega^2 C_\theta + \sigma_\epsilon^2 I_d, \quad (\text{S1})$$

$$y_t = G y_{t-1} + \eta_t, \quad \eta_t \sim N(0, \Sigma_\eta), \quad y_0 \sim N(m_0, C_0), \quad (\text{S2})$$

for $t = 1, \dots, T$, with z_t the d observations at time t , A the $d \times p$ loading matrix and C_0 known; in the paper $p = 1$ and $A = \mathbf{1}_d$. The package writes $\Sigma_e = \sigma_\omega^2 \Gamma$ with $\Gamma = C_\theta + \gamma I_d$ and $\gamma = \sigma_\epsilon^2 / \sigma_\omega^2$, and estimates $\psi = (\beta, \sigma_\omega^2, \log \gamma, \log \theta, G, \Sigma_\eta, m_0)$ by the EM algorithm of [Shumway and Stoffer \(1982\)](#): the E-step is the Kalman smoother, the M-step updates $G, \Sigma_\eta, m_0, \sigma_\omega^2$ and β in closed form and $(\log \theta, \log \gamma)$ by Newton–Raphson. The log-likelihood $\ell(\psi)$ is the by-product of the Kalman filter (the prediction-error decomposition).

Treating the states $y_{0:T}$ as missing data, the complete-data log-likelihood is, up to a constant,

$$\begin{aligned} \ell_c(\psi) = & -\frac{1}{2} \log |C_0| - \frac{1}{2} (y_0 - m_0)^\top C_0^{-1} (y_0 - m_0) \\ & - \frac{T}{2} \log |\Sigma_\eta| - \frac{1}{2} \sum_{t=1}^T (y_t - G y_{t-1})^\top \Sigma_\eta^{-1} (y_t - G y_{t-1}) \\ & - \frac{T}{2} \log |\Sigma_e| - \frac{1}{2} \sum_{t=1}^T (z_t - X_t \beta - A y_t)^\top \Sigma_e^{-1} (z_t - X_t \beta - A y_t), \end{aligned} \quad (\text{S3})$$

the sum of three terms, for the initial state, the state equation and the measurement equation, each involving its own parameters. The EM algorithm ([Dempster et al., 1977](#)) maximizes the observed log-likelihood $\ell(\psi)$ by iterating an E-step, which computes $Q(\psi \mid \psi^{(j)}) = \mathbb{E}\{\ell_c(\psi) \mid z_{1:T}; \psi^{(j)}\}$, and an M-step, which increases Q over ψ .

S1.2 The Kalman filter and smoother

Every algorithm of the package rests on three recursions, run at given parameters by the function `kalman()`.

The filter. Starting from $y_{0|0} = m_0$ and $P_{0|0} = C_0$, for $t = 1, \dots, T$,

$$\begin{aligned} y_{t|t-1} &= G y_{t-1|t-1}, & P_{t|t-1} &= G P_{t-1|t-1} G^\top + \Sigma_\eta, \\ v_t &= z_t - X_t \beta - A y_{t|t-1}, & F_t &= A P_{t|t-1} A^\top + \Sigma_e, \\ y_{t|t} &= y_{t|t-1} + K_t v_t, & P_{t|t} &= P_{t|t-1} - K_t A P_{t|t-1}, & K_t &= P_{t|t-1} A^\top F_t^{-1}, \end{aligned}$$

and the log-likelihood follows from the prediction-error decomposition,

$$\ell(\psi) = -\frac{1}{2} \sum_{t=1}^T \{d \log(2\pi) + \log |F_t| + v_t^\top F_t^{-1} v_t\}. \quad (\text{S4})$$

The $d \times d$ matrix F_t is never formed. Σ_e does not depend on t and $A P_{t|t-1} A^\top$ has rank p , so the Woodbury identity and the matrix determinant lemma give

$$\begin{aligned} F_t^{-1} &= \Sigma_e^{-1} - \Sigma_e^{-1} A (P_{t|t-1}^{-1} + A^\top \Sigma_e^{-1} A)^{-1} A^\top \Sigma_e^{-1}, \\ \log |F_t| &= \log |\Sigma_e| + \log |P_{t|t-1}| + \log |P_{t|t-1}^{-1} + A^\top \Sigma_e^{-1} A|, \end{aligned}$$

and the Cholesky factor of Σ_e , $\Sigma_e^{-1}A$ and $A^\top \Sigma_e^{-1}A$ are computed once per pass: the cost of a pass is $O(d^3 + Td^2p)$ instead of $O(Td^3)$. The log-density is evaluated directly, since on a few hundred locations the density itself underflows. When some responses are missing at time t , the rows of the measurement equation that are not observed are dropped (Durbin and Koopman, 2012, Sect. 4.10), and the constants above are cached by pattern of missingness.

The smoother. With the gains $J_t = P_{t|t}G^\top P_{t+1|t}^{-1}$, the fixed-interval smoother runs backwards, for $t = T-1, \dots, 0$,

$$y_{t|T} = y_{t|t} + J_t(y_{t+1|T} - y_{t+1|t}), \quad P_{t|T} = P_{t|t} + J_t(P_{t+1|T} - P_{t+1|t})J_t^\top,$$

the last step giving the smoothed initial state $y_{0|T}$, $P_{0|T}$, with $J_0 = C_0G^\top P_{1|0}^{-1}$.

The lag-one covariance smoother. The covariances $P_{t,t-1|T} = \text{Cov}(y_t, y_{t-1} \mid z_{1:T})$ follow the backward recursion of Shumway and Stoffer (1982):

$$\begin{aligned} P_{T,T-1|T} &= (I - K_TA)G P_{T-1|T-1}, \\ P_{t-1,t-2|T} &= P_{t-1|t-1}J_{t-2}^\top + J_{t-1}(P_{t,t-1|T} - GP_{t-1|t-1})J_{t-2}^\top, \quad t = T, \dots, 2. \end{aligned}$$

With missing responses at time T , K_T is built on the observed rows, and it is zero when nothing is observed.

S1.3 The EM algorithm for STEM

E-step. Taking the expectation of (S3) given the data, Q depends on the states only through the smoothed initial state and

$$\begin{aligned} S_{11} &= \sum_{t=1}^T (y_{t|T} y_{t|T}^\top + P_{t|T}), & S_{00} &= \sum_{t=1}^T (y_{t-1|T} y_{t-1|T}^\top + P_{t-1|T}), \\ S_{10} &= \sum_{t=1}^T (y_{t|T} y_{t-1|T}^\top + P_{t,t-1|T}), & W &= \sum_{t=1}^T (AP_{t|T}A^\top + r_t r_t^\top), \quad r_t = z_t - X_t \beta^{(j)} - A y_{t|T}, \end{aligned}$$

so that

$$\begin{aligned} -2Q(\psi \mid \psi^{(j)}) &= \log |C_0| + \text{tr}[C_0^{-1}\{P_{0|T} + (y_{0|T} - m_0)(y_{0|T} - m_0)^\top\}] \\ &\quad + T \log |\Sigma_\eta| + \text{tr}[\Sigma_\eta^{-1}(S_{11} - S_{10}G^\top - GS_{10}^\top + GS_{00}G^\top)] \\ &\quad + T \log |\Sigma_e| + \text{tr}(\Sigma_e^{-1}W), \end{aligned} \tag{S5}$$

up to a constant and with β entering the last term through W . With missing responses, W and the response in the update of β are completed by their conditional expectations given the observed data (Section S2). In the package the E-step is `stem_estep()`, which calls `kalman()`.

M-step. The parameters are updated in turn, each given the updates that precede it (an ECM algorithm, Meng and Rubin, 1993): the initial state, $m_0 \leftarrow y_{0|T}$; the state equation, with Σ_η and G diagonal (the default),

$$[\Sigma_\eta]_{ii} \leftarrow \frac{1}{T} [S_{11} - S_{10}G^\top - GS_{10}^\top + GS_{00}G^\top]_{ii}, \quad G_{ii} \leftarrow \frac{[\Sigma_\eta^{-1}S_{10}]_{ii}}{[\Sigma_\eta^{-1}S_{00}]_{ii}},$$

and otherwise $\Sigma_\eta \leftarrow T^{-1}(S_{11} - S_{10}S_{00}^{-1}S_{10}^\top)$ and $G \leftarrow S_{10}S_{00}^{-1}$; the scale of the measurement error, $\sigma_\omega^2 \leftarrow \text{tr}(\Gamma^{-1}W)/(dT)$, at the current (θ, γ) ; the regression coefficients, by generalized least squares given the smoothed states,

$$\beta \leftarrow \left(\sum_{t=1}^T X_t^\top \Sigma_e^{-1} X_t \right)^{-1} \sum_{t=1}^T X_t^\top \Sigma_e^{-1} (\hat{z}_t - A y_{t|T}),$$

with $\hat{z}_t$ the completed response and $\Sigma_e = \sigma_\omega^2 \Gamma$ at the updated σ_ω^2 (with a ridge or an elastic net, the corresponding penalized quadratic, Section S4); and the spatial parameters, by Newton–Raphson on $(\log \theta, \log \gamma)$ for

$$Q_e(\theta, \gamma) = T \log |\sigma_\omega^2 \Gamma| + \text{tr}\{(\sigma_\omega^2 \Gamma)^{-1} W\},$$

minus twice the part of Q that depends on them, safeguarded as described in Section S1.10. Finally $\sigma_\epsilon^2 = \gamma \sigma_\omega^2$. Every update maximizes Q in its own block or, for the Newton step, does not decrease it, so the algorithm is a generalized EM algorithm and ℓ does not decrease along the iterations. In the package the M-step is `stem_mstep_em()`, built from the updates of `R/mstep-updates.R`.

S1.4 The EM algorithm within SC-STEM

SC-STEM fits one STEM model per regime, on the locations $\mathcal{C}_k$ with d_k , the rows A_k of the loading matrix and the parameters ψ_k , and alternates between the parameters and the labels (Algorithm 1 of the manuscript). The EM algorithm enters in two places.

1. In the parameter step of the alternation, where it only has to rank the regimes for the label step: a few plain EM (or ECME) iterations per regime, with tolerances 10^{-2} and 1 and at most 50 iterations, without acceleration. The first time, a regime starts from the least-squares coefficients of its own locations (an unpenalized fit) or from the final refit of the unpenalized solution the fit starts from (a penalized one); every later time, from its own estimates of the previous iteration. The label step scores each location with the smoothed states and the measurement parameters of each regime.
2. In the final refit, where the regimes are re-estimated once on the final partition, with the settings of the pooled fit and the algorithm chosen, starting from the estimates of the last iteration of the alternation. The reported estimates and log-likelihoods come from this refit. Fits that end at the same partition share one refit (Section 4.1 of the manuscript).

S1.5 Where the EM algorithm is slow

The EM algorithm is a fixed-point iteration $\psi^{(j+1)} = M(\psi^{(j)})$. Near the maximum $\hat{\psi}$ it is linear,

$$\psi^{(j+1)} - \hat{\psi} \approx J(\psi^{(j)} - \hat{\psi}), \quad J = I_c^{-1} I_m,$$

with I_c the complete-data information and $I_m = I_c - I_o$ the information that is missing because the states are not observed (Dempster et al., 1977; Meng and Rubin, 1994). The eigenvalues of J lie in $[0, 1)$, and convergence is linear with rate λ , the largest of them: reducing the distance from the maximum by a factor ε takes about $\log \varepsilon / \log \lambda$ iterations, 95 at $\lambda = 0.93$ and 572 at $\lambda = 0.988$ for $\varepsilon = 10^{-3}$. A stopping rule that looks at the change between two iterations leaves, along a direction with rate λ ,

$$\|\psi^{(j)} - \hat{\psi}\| \approx \frac{\|\psi^{(j+1)} - \psi^{(j)}\|}{1 - \lambda}, \quad \ell(\hat{\psi}) - \ell(\psi^{(j)}) \approx \frac{\ell(\psi^{(j+1)}) - \ell(\psi^{(j)})}{1 - \lambda^2},$$

so that a tolerance τ on the relative change of the parameters leaves a relative error of about $\tau/(1 - \lambda)$: the same tolerance means different accuracies in different regimes, and a slow direction is where the rule stops early.

On the regimes of SC-STEM three situations make λ close to one or the objective flat.

1. *The split of the common variation between the latent process and the field.* The latent process is common to the locations of a regime; the field, with a range comparable to the size of the regime, is nearly so at every time. Only their dynamics tell them apart, the latent process being persistent and the field white in time, and the unobserved states carry most of the information about the split. Along $(\sigma_\omega^2, \sigma_\epsilon^2, \theta, \Sigma_\eta)$ the rate is about 0.93 in the regimes of the simulation study (Table S1); this direction sets the number of iterations.
2. *The intercept and the initial level of the latent process.* The mean of the data is $X_t\beta + AG^tm_0$: a change of m_0 moves the first periods, which the intercept and the smoothed states partly absorb. The E-step attributes a level error mostly to the smoothed states, the M-step then updates m_0 and β given them and corrects only a small part of the error at every iteration. The rate is about 0.988, the direction carries little likelihood, and it is where the error at the stop sits. When G is close to zero, m_0 enters the likelihood only through Gm_0 , and G and m_0 trade along a ridge.
3. *The range at the boundary of what the distances identify.* On a small set of locations the likelihood is flat in θ beyond two limits (Section S1.10); the Newton–Raphson step on a flat objective took steps of any size, and every later iteration paid repeated grid searches.

Table S1: Rate (ratio of two successive steps) and direction of the iterations on regime 1 of replication 1 of the reference cell of the simulation study, from its least-squares start. Direction: the step of each parameter relative to its value, scaled to unit norm. The EM algorithm moves along the split of the variance up to about iteration 50 and along the direction of the intercept and m_0 from about iteration 80; ECME (Section S1.6) moves along the first throughout.

	rate	β_0	β_1	m_0	G	σ_η^2	σ_ω^2	σ_ϵ^2	θ
EM, iteration 30	0.938	−0.02	0.01	0.04	0.03	−0.17	0.88	−0.40	0.18
EM, iteration 200	0.988	−0.11	0.00	0.99	0.01	−0.01	0.00	0.00	0.00
ECME, iteration 30	0.938	0.00	0.01	−0.15	0.03	−0.17	0.87	−0.40	0.17
ECME, iteration 200	0.929	0.00	0.01	−0.15	0.04	−0.19	0.87	−0.41	0.15

S1.6 ECME

The ECME algorithm, “expectation / conditional maximization either” (Liu and Rubin, 1994), lets some of the conditional maximizations of the M-step maximize the observed log-likelihood ℓ instead of Q , under the same constraint. Along the directions those steps move, the algorithm no longer pays the fraction of missing information. It remains monotone when the steps on ℓ follow those on Q within an iteration (otherwise the E-step has to be recomputed in between, as in the AECEM algorithm of Meng and van Dyk, 1997).

The package updates $\delta = (\beta^\top, m_0^\top)^\top$ in this way. An iteration runs the E-step, updates Σ_η , G , σ_ω^2 and (θ, γ) on Q exactly as the EM algorithm does, and then maximizes ℓ over δ given all of them. Given the other parameters, ℓ depends on δ only through the mean of the data, $E(z_t) = X_t\beta + AG^tm_0$, which is linear in δ : ℓ is a quadratic in δ and its maximizer is a generalized least-squares estimator

under the covariance the model implies, latent process included. The augmented Kalman filter computes it without forming that $dT \times dT$ covariance (de Jong, 1991); see also Durbin and Koopman (2012, Ch. 6). The filter of the current parameters is run on every column of the $d \times (1 + r + p)$ matrix

$$\begin{bmatrix} z_t, & X_t, & AG^t \end{bmatrix},$$

the first column with the current m_0 as initial mean and the others with zero, the last block standing for a change Δm_0 of m_0 . The gains do not depend on the data, so all the columns share them and share F_t ; and the filter is linear in its data and in its initial mean, so with $\begin{bmatrix} e_t, & E_t \end{bmatrix}$ the innovations of the columns, the innovations of z_t at $(\beta, m_0 + \Delta m_0)$ are $e_t - E_t(\beta^\top, \Delta m_0^\top)^\top$. Hence

$$\ell = c - \frac{1}{2} \sum_{t=1}^T (e_t - E_t \delta)^\top F_t^{-1} (e_t - E_t \delta), \quad \hat{\delta} = M^{-1} v, \quad M = \sum_{t=1}^T E_t^\top F_t^{-1} E_t, \quad v = \sum_{t=1}^T E_t^\top F_t^{-1} e_t, \quad (\text{S6})$$

with F_t^{-1} applied through the Woodbury identity and the missing rows dropped, as in the filter. With a penalty on β , m_0 is profiled out, $M_{\beta\beta} = M_{\beta\beta} - M_{\beta m} M_{mm}^{-1} M_{m\beta}$ and $v_{\beta\beta} = v_\beta - M_{\beta m} M_{mm}^{-1} v_m$, the quadratic in β is penalized as in the EM algorithm, and $\Delta m_0 = M_{mm}^{-1} (v_m - M_{m\beta} \beta)$. Without a latent process there is no m_0 column. An ECME iteration costs one more filter pass than an EM iteration, about 25% more time.

The step removes the second slow direction of Section S1.5 at once (Table S1: the late iterations of ECME no longer move β_0 and m_0), and with it the ridge between G and m_0 ; it does not change the first. In the package the M-step is `stem_mstep_ecme()`, and the augmented filter `stem_gls_mean()`.

S1.7 SQUAREM

SQUAREM (Roland and Varadhan, 2005; Varadhan and Roland, 2008) accelerates any fixed-point iteration $\psi \leftarrow M(\psi)$, here an EM or an ECME iteration, by extrapolating from two of its steps. Let $u(\psi)$ be the vector of the free parameters on a scale on which every value is admissible,

$$u(\psi) = (\beta, m_0, \text{vec } G, \log \text{diag } \Sigma_\eta, \log \sigma_\omega^2, \log \gamma, \log \theta)$$

(the logarithm of the diagonal of the Cholesky factor of Σ_η and its off-diagonal elements when Σ_η is not diagonal). A cycle starts from ψ_0 and computes $\psi_1 = M(\psi_0)$ and $\psi_2 = M(\psi_1)$, then, with $u_i = u(\psi_i)$,

$$r = u_1 - u_0, \quad w = u_2 - 2u_1 + u_0, \quad \alpha = \min \left\{ \alpha_{\max}, \max \left(1, \frac{\|r\|}{\|w\|} \right) \right\}, \quad u' = u_0 + 2\alpha r + \alpha^2 w, \quad (\text{S7})$$

the scheme S3 of Varadhan and Roland (2008). For a linear iteration in one direction with rate λ , $r = (\lambda - 1)e$ and $w = (\lambda - 1)^2 e$ with $e = u_0 - \hat{u}$, so $\|r\|/\|w\| = 1/(1 - \lambda)$ and $u' - \hat{u} = \{1 - \alpha(1 - \lambda)\}^2 e = 0$: the extrapolation removes that direction exactly, and in several directions it removes most of the slowest. With $\alpha = 1$, $u' = u_2$, two plain steps. When $\alpha > 1$ one more iteration is run from ψ' , which gives $\ell(\psi')$ and stabilizes the extrapolation: the cycle ends at $M(\psi')$ if $\ell(\psi') \geq \ell(\psi_0)$, and at ψ_2 otherwise, whose increase the EM algorithm guarantees. The safeguard costs nothing, since every iteration evaluates ℓ at its input. The bound $\alpha_{\max}$ starts at 1, grows by a factor 4 every time it binds and shrinks by the same factor after a rejection, as in the R package SQUAREM (Du and Varadhan, 2020). The extrapolated θ is kept within the limits of Section S1.10.

The stopping rule (Section S1.9) is applied at the start of every cycle, to the step from ψ_0 to ψ_1 , and the number of iterations reported counts every call of M . SQUAREM is not used in two cases. With a penalty on β the objective is the penalized likelihood, whose scale is re-measured

at every iteration, so the safeguard has no fixed objective; the plain iterations are run. Inside the alternation of SC-STEM the fits stop after a few iterations, where the cycles cost more than they save, and extrapolations from points far from convergence were seen to change the partitions the alternation reaches; the plain iterations are run there too. In the package the iterations are `stem_iterate_squarem()`.

S1.8 Implementation in the package

The algorithm is chosen by `STEM_control(algorithm = )`: "SQUAREM" (the default), "EM", "ECME" or "SQUAREM-ECME". The code is organized by role (Table S2): one iteration is a *map*, the parameters in and the parameters out together with the log-likelihood at the parameters in, and the map is run either plainly or by SQUAREM until the stopping rule. `STEM_Estimation()` builds the data of the fit once (distances, covariates, the index of the observed rows), runs the algorithm, and calls `kalman()` once more at the estimates for the log-likelihood and the smoothed states it reports, so that estimates, log-likelihood and states always refer to the same parameters.

Table S2: The estimation code of `Stem`, by file.

file	content
R/kalman.R	<code>kalman()</code> : the filter (<code>filtering()</code> , <code>filterstep()</code>), the smoother (<code>smoothing()</code>), the smoothed initial state and the lag-one covariances
R/estep.R	<code>stem_estep()</code> : the expected sufficient statistics S_{11} , S_{10} , S_{00} , W and the completed response
R/mstep-updates.R	the conditional updates: latent process, σ_ω^2 , β given the smoothed states, the spatial parameters by safeguarded Newton–Raphson, the limits of θ
R/mstep-em.R, R/mstep-ecme.R	the M-steps of EM and ECME; the augmented filter of ECME, <code>stem_gls_mean()</code>
R/squarem.R	SQUAREM: <code>stem_iterate_squarem()</code> and the unconstrained parametrization
R/em-fit.R	the wrapper <code>stem_em_fit()</code> , the plain iterations, the stopping rule

The settings are those of `STEM_control()`: `em_tol_par`, `em_tol_loglik`, `em_maxit` and `em_stop` for the pooled fit and the final refits; `alt_em_tol_par`, `alt_em_tol_loglik` and `alt_em_maxit` inside the alternation; `nr_maxit` for the Newton–Raphson step. Each fit reports in `convergence.par` the number of iterations, the Newton–Raphson iterations of each, the algorithm, whether the stopping rule was met, the limits of θ and whether the estimate sits at one of them.

S1.9 Stopping rule, and the rule of D-STEM

Two criteria are computed at every iteration (at every cycle under SQUAREM): the largest relative change of a free parameter, taken one at a time,

$$\max_l \frac{|\psi_l^{(j+1)} - \psi_l^{(j)}|}{\max(|\psi_l^{(j)}|, 10^{-3})},$$

on the scale on which the algorithm updates it and with the loading matrix and C_0 excluded, and the absolute change of the log-likelihood, $|\ell(\psi^{(j+1)}) - \ell(\psi^{(j)})|$. The algorithm stops as soon as either falls below its tolerance, by default 10^{-3} for both, or after 500 iterations; requiring both at the same iteration is an option. Table S3 compares the rule with that of D-STEM v2 (Wang et al., 2021).

Table S3: The stopping rules of **Stem** and of D-STEM v2. With log-likelihoods between -4000 (a regime) and -15000 (a pooled fit) the relative tolerance of D-STEM on ℓ is 0.4 to 1.5 units.

	Stem	D-STEM v2
parameters	$\max_l \Delta\psi_l / \max(\psi_l , 10^{-3}) < 10^{-3}$	$\max_l \Delta\psi_l / \psi_l < 10^{-4}$
log-likelihood	$ \Delta\ell < 10^{-3}$ (absolute)	$ \Delta\ell / \ell < 10^{-4}$ (relative)
combination	either criterion	either criterion
iterations	at most 500 (50 in the alternation)	at most 100

D-STEM is the stricter on the parameters and the far looser on the log-likelihood: its relative tolerance is 400 to 1500 times the absolute tolerance of **Stem** on these fits. Since either criterion stops the iterations, the one on ℓ is met first in D-STEM, and the iterations stop while several units of log-likelihood can still be gained; with the rule of **Stem** and SQUAREM the distance from the maximum at the stop is of the order of 10^{-3} units (Section S1.13). The criterion on ℓ is absolute because the differences that matter, between fits and in the information criteria, are absolute, whereas a relative criterion loosens as the log-likelihood grows with the size of the data. Tolerances of 10^{-3} on both criteria left the estimates within 1 to 2% of those at 10^{-4} and the selected models unchanged in the simulation study, while 10^{-2} moved the range and the variances by tens of percent.

S1.10 The limits of the range

The distances between the locations identify the range only within limits. When the correlation between the two closest locations falls below 0.05, the field is white within the set of locations and cannot be told from the nugget; when the correlation between the two farthest exceeds 0.95, the field is constant over the locations at every time and cannot be told from an effect common to them, the latent process included. Beyond either limit Q_e and ℓ are flat in θ . The package therefore keeps

$$\theta_{\min} = \frac{-\log 0.95}{h_{\max}} \leq \theta \leq \theta_{\max} = \frac{-\log 0.05}{h_{\min}}, \quad (\text{S8})$$

with $h_{\min}$ and $h_{\max}$ the smallest and largest distance between two locations of the fit (of the regime, in SC-STEM), in the Newton–Raphson step and in the extrapolation of SQUAREM. The Newton–Raphson step on $(\log \theta, \log \gamma)$ is safeguarded in two further ways. A Newton step $-H^{-1}g$ is taken only when the Hessian H of Q_e is positive definite ($H_{11} > 0$, $\det H > 10^{-3}$), and is accepted only if it does not increase Q_e , halving it otherwise up to ten times. When H is not positive definite, the point moves to the best of a grid of ten by ten values, from 0.01 to 10 times the current θ and γ within the limits, if that lowers Q_e , and the iterations resume from there; a second Hessian that is not positive definite ends them, since Q_e is then flat. A fit whose range ends at a limit reports it, as `convergence.par$theta_bound` in `STEM_Estimation()` and `theta_bound` in `SCSTEM_Estimation()`; the range is then not estimated freely, though it is still counted among the parameters of the information criteria.

Without the limits and the safeguards, the Newton–Raphson step on a flat Q_e took steps of any size (θ was seen to move from 0.02 to e^{-700} in a single M-step), every later iteration paid up to thirty grid searches of a hundred evaluations each, and SQUAREM, extrapolating along the flat direction, ran to the limit on the iterations. Where neither the limits nor the safeguards come into play, the iterations are those of the unconstrained algorithm (Section S1.13).

S1.11 No ridge in the inversions

The earlier versions of the package added $0.01 I$ to every matrix the iterations inverted, while the filter that computes ℓ added nothing. The iterations then converge to a fixed point ψ_r that is not $\hat{\psi}$: a perturbation δ of the map moves its fixed point by $(I - J)^{-1}\delta$, so a perturbation of 1% of the update of σ_ω^2 is amplified along the split of the variance by $1/(1 - 0.93) \approx 14$. On replication 1 of the reference cell the variance parameters were off by 3 to 13% in the regimes and Σ_η of the pooled fit by 84%, at any tolerance (Table S4). Every matrix inverted is positive definite at admissible parameters, and no ridge is added by default (`regularization = 0`).

Table S4: The fixed point of the iterations with a ridge of 0.01 against the maximum of the likelihood (BFGS on the exact log-likelihood), replication 1 of the reference cell: shortfall of the log-likelihood and relative difference of the parameters (%). Without the ridge the same iterations converge to the maximum.

fit	d	$\ell(\hat{\psi}) - \ell(\psi_r)$	G	σ_η^2	σ_ω^2	σ_ϵ^2	θ
regime 1	34	0.43	-0.3	+1.6	-7.6	+3.8	-1.7
regime 2	33	0.31	-0.3	+0.3	-9.6	+3.1	-4.7
regime 3	33	0.52	+77.9	+0.0	-12.9	+6.9	-9.1
pooled	100	1.73	-6.4	+84.3	-13.3	+0.3	+13.4

S1.12 The four algorithms at a glance

The four algorithms maximize the same likelihood and share the same E-step: the Kalman filter and smoother compute the exact conditional moments of the latent states given the data, which in a linear Gaussian state-space model is exact posterior inference. They differ in what the M-step maximizes and in how successive iterations are combined.

- **EM** maximizes the expected complete-data log-likelihood block by block. In the language of machine learning it is coordinate ascent on the evidence lower bound, made tight by the E-step. It is monotone and simple, and it converges linearly, slowly wherever the unobserved states carry most of the information about a direction of the parameters.
- **ECME** updates the regression coefficients and the initial level of the latent process on the observed log-likelihood, with the states integrated out (a “collapsed” update, computed as a generalized least-squares step by the Kalman filter run on the covariates). It removes the slow direction in which the intercept and the initial level move together, at about 25% more time per iteration, and leaves the other directions as they are.
- **SQUAREM** extrapolates two successive EM iterations along the direction in which they converge most slowly, an extrapolation of the fixed-point map in the family of Aitken’s and Anderson’s accelerations, with a monotonicity safeguard that rejects any extrapolation that would lower the likelihood. It changes neither the E-step nor the M-step.
- **SQUAREM-ECME** extrapolates ECME iterations in the same way.

Table S5 summarizes their behaviour on the simulation study (Section S1.13). SQUAREM is the default of the package: it took about a third of the time of EM on single fits and about half on whole SC-STEM grids, stopped closer to the maximum, and selected the same models.

Table S5: The four algorithms. Time and iterations relative to EM on whole SC-STEM grids (six cells of the simulation study, two replications each); shortfall: median shortfall of the log-likelihood at the stop on single fits. All at the default tolerances. All four are monotone; the two accelerated ones through a safeguard on the extrapolation.

algorithm	M-step	iterations combined	time	iterations	shortfall
EM	conditional maximizations of Q	plain	1	1	0.020
ECME	as EM, with (β, m_0) on ℓ	plain	1.17	0.96	0.017
SQUAREM	as EM	two extrapolated	0.53	0.41	0.0027
SQUAREM-ECME	as ECME	two extrapolated	0.62	0.39	0.0023

S1.13 Numerical experiments

The experiments use the data-generating process of the simulation study (Supplementary Material B). Single fits are those of the three regimes of the reference cell (S1s-ind, $n = 100$ locations, $T = 120$, $\omega = 0.7$; regimes of 33 or 34 locations), each on its true partition and from the least-squares start of its own locations, as a final refit would be, and of the pooled model on the 100 locations; whole fits are the grid of $K \in \{1, \dots, 4\}$ and seven values of ϕ followed by the selection rule, on six cells chosen among the lightest and the heaviest of the design. The scripts are in the replication material.

Correctness of the implementation. With the plain EM algorithm and a ridge of 0.01, the code organized as in Table S2 reproduces the previous implementation of the package to 10^{-13} , iteration by iteration and over whole fits, on complete and incomplete data, with the latent process or the spatial correlation switched off, under a ridge and an elastic net on β , and with $p = 2$. The augmented filter of ECME reproduces, to 10^{-10} , the generalized least-squares estimator computed on the explicit $dT \times dT$ covariance of the data, missing values included. Without the ridge, the iterations converge to the maximum of the exact log-likelihood found by BFGS (Table S4).

Exactness of what the package reports. The checks above compare the package with itself or with the filter. The quantities it reports were also compared with computations on the full covariance of the stacked observations, on a model of seven locations and thirty periods with twelve values missing: the log-likelihood returned is the Gaussian log-likelihood of the observed values to 10^{-14} , under EM, ECME and SQUAREM; the estimates are the maximum of that log-likelihood found by a general-purpose optimizer, to a largest relative difference of $2 \cdot 10^{-5}$; the smoothed states and the fitted signal are the conditional means to $3 \cdot 10^{-15}$. The simulator used by both bootstraps reproduces the mean and the covariance of the model within Monte Carlo error over 20,000 data sets. These checks are part of the test suite of the package. They found one defect, inherited from the first version of the package: the log-likelihood returned by the pooled fit was four times the log-likelihood, since the EM loop stored -2ℓ and multiplied it by -2 again. The iterations were unaffected, but the information criteria computed from the returned value weighted the likelihood four times against the penalty; the version described here returns ℓ , and every result in the paper and in the supplements uses it.

The four algorithms reach the same maximum. On a small simulated model run to a tolerance of 10^{-7} on both criteria, the four algorithms stop at the same log-likelihood to 10^{-9} and the same range to 10^{-6} , after 177 (EM), 178 (ECME), 39 (SQUAREM) and 30 (SQUAREM-ECME) iterations. On the single fits of ten replications, the maximum was computed by SQUAREM-ECME

at a tolerance of 10^{-10} , and it coincides with the BFGS maximum wherever the latter was computed. At a given tolerance the algorithms stop at different distances from it (Table S6): at the default tolerances the median shortfall of the log-likelihood is 0.02 units for EM and ECME and 0.003 for SQUAREM and SQUAREM-ECME, and the median of the largest relative distance of a parameter from the maximum is 3.8%, 3.2%, 1.4% and 1.1%. The largest distances are those of parameters near zero (the persistence of a regime with $G \approx 0.02$, the innovation variance of a pooled fit with $\Sigma_\eta \approx 0$), whose relative error is large while the absolute one is small.

Table S6: Single fits (three regimes and the pooled model, ten replications of the reference cell) at three tolerances on both criteria. Time: seconds per replication (four fits); iterations: mean per fit; shortfall: $\ell(\hat{\psi}) - \ell$ at the stop, median and maximum; error: largest relative distance of a parameter other than m_0 from the maximum, median and 90th percentile (%).

tolerance	algorithm	time	iterations	shortfall		error	
				median	max	median	p90
10^{-2}	EM	5.5	38.5	0.12	18	8.0	97
	ECME	6.5	36.2	0.11	18	7.2	92
	SQUAREM	2.4	18.1	0.065	18	6.2	92
	SQUAREM-ECME	2.8	15.8	0.10	18	7.3	69
10^{-3} (default)	EM	12.7	89.0	0.020	4.2	3.8	28
	ECME	15.2	83.5	0.017	4.3	3.2	20
	SQUAREM	4.6	34.3	0.0027	0.17	1.4	9.6
	SQUAREM-ECME	5.2	29.6	0.0023	4.3	1.1	6.5
10^{-4}	EM	23.6	165.7	0.0026	0.23	1.3	10
	ECME	26.7	148.8	0.0020	0.23	1.1	6.3
	SQUAREM	6.7	50.2	0.00028	0.053	0.38	4.5
	SQUAREM-ECME	7.6	42.8	0.000032	0.050	0.10	1.2

The maxima of the shortfall at 10^{-3} come from one regime, where EM, ECME and SQUAREM-ECME stopped on a plateau on which the field takes the place of the latent process ($\theta = 0.21$ against 2.45 at the maximum, 4 units short) and SQUAREM, extrapolating, crossed it.

Accuracy against the true values. The numerical error adds to the statistical one. Table S7 gives the relative root mean squared error of the estimates of the three regimes against the true values of the simulation: at the default tolerances SQUAREM matches the exact maximum on every parameter, while EM, ECME and SQUAREM-ECME, which stop farther from it, lose accuracy on the range (24.5% to 24.9% against 18.5%) and on σ_ϵ^2 . At 10^{-4} every algorithm is within 0.3 points of the maximum on every parameter. The regression coefficients are unaffected throughout.

Computing time on whole fits. On a machine with cores of different speed, wall times are comparable only within one process. Each job of this experiment ran the whole grid and the selection under the four algorithms one after the other, in the same process and in an order rotated across jobs, and the time relative to EM was estimated net of the position in the job by a regression of the log-time on the job, the algorithm and the position (Table S8). SQUAREM took 0.53 times the time of EM (95% interval 0.45 to 0.64), 0.58 on the light cells and 0.46 on the heavy ones; SQUAREM-ECME 0.62; ECME 1.17. No fit reached the limit on the iterations, every algorithm selected the same number of regimes in every job, and at the true number of regimes SQUAREM

Table S7: Relative root mean squared error (%) against the true values, three regimes of the reference cell, ten replications, at the default tolerances. “Maximum”: the exact maximum of the likelihood.

parameter	maximum	EM	ECME	SQUAREM	SQUAREM-ECME
β_0	5.3	5.4	5.3	5.4	5.3
β_1	2.1	2.1	2.1	2.1	2.1
G	34.2	31.1	33.6	32.1	33.6
σ_η^2	16.4	17.0	17.2	16.4	17.1
σ_ϵ^2	6.8	8.9	8.9	6.8	9.3
σ_ω^2	12.3	10.7	10.7	12.1	11.7
θ	18.5	24.5	24.6	18.8	24.9

was at most 0.10 units of log-likelihood from the best of the four. EM was up to 3.4 units short, ECME and SQUAREM-ECME up to 10.9, in the weak scenario at $n = 40$.

Table S8: Whole SC-STEM grids, two replications per cell: time relative to EM (geometric mean over the replications) and seconds of EM per replication.

cell	EM (s)	ECME	SQUAREM	SQUAREM-ECME
S1s-ind, $n = 40$, $T = 60$	21	1.12	0.75	0.90
S1w-shr, $n = 40$, $T = 60$	30	1.36	0.53	0.74
S1s-ind, $n = 100$, $T = 120$	68	1.17	0.49	0.55
S3Seta-shr, $n = 200$, $T = 365$	239	1.25	0.51	0.59
S1s-ind, $n = 200$, $T = 365$, $\omega = 0$	399	1.15	0.30	0.35
S2, $n = 400$, $T = 120$	293	1.02	0.78	0.78
all, net of the position		1.17	0.53	0.62

The limits of the range. The same grids were run before and after the limits and the safeguards of Section S1.10. On the 24 single fits of three replications of the reference cell, where nothing binds, the results are identical to 10^{-9} , iterations included. On the whole grids, the fits at the true number of regimes are unchanged in 40 of 48 cases and better in the other 8, all in the weak scenario at $n = 40$, by 0.3 to 16.5 units of log-likelihood; the selected models are unchanged. The total time of the grids fell from 2585 to 2102 seconds for EM and from 1747 to 1057 for SQUAREM; on the replication that had cost most, a small cell whose regimes ran towards the boundary, from 362 to 29 seconds for EM and from 1416 to 25 for SQUAREM, where two regimes had reached the limit of 500 iterations.

S2 Missing responses: the completed sufficient statistics

Section 4.6 of the manuscript states the form of the expected complete-data sufficient statistic of the measurement equation when part of the response is unobserved. This section derives it. Throughout, the regime index k is fixed and suppressed, A denotes the $d \times p$ loading matrix, Σ_e the measurement covariance, and $\mathcal{Z}$ the whole of the observed data.

S2.1 Setting

The measurement equation is $z_t = X_t\beta + Ay_t + e_t$ with $e_t \sim N(0, \Sigma_e)$, independent across t and of the latent process. Partition the locations at time t into the observed set O and the missing set M , and write the corresponding blocks of any vector or matrix with those subscripts. The EM algorithm treats both the latent states *and* the unobserved responses as missing data, so the E-step must return

$$\mathbb{E}[e_t e_t^\top | \mathcal{Z}], \quad \text{and} \quad \mathbb{E}[z_t | \mathcal{Z}], \quad (\text{S9})$$

the first entering the updates of the variance components and of the spatial covariance parameters, the second the update of β .

The Kalman smoother, run on the selected measurement equation of [Durbin and Koopman \(2012, Sect. 4.10\)](#), supplies

$$\hat{y}_t = \mathbb{E}[y_t | \mathcal{Z}], \quad C_t^s = \text{Var}[y_t | \mathcal{Z}]. \quad (\text{S10})$$

S2.2 The conditional law of the measurement error

Conditionally on the latent state y_t , the measurement error is $e_t = z_t - X_t\beta - Ay_t$, so its observed block $e_{O,t}$ is a known linear function of y_t once $z_{O,t}$ is observed, whereas its missing block is not observed at all. Because e_t is Gaussian with covariance Σ_e , the standard conditioning formula gives

$$e_{M,t} | e_{O,t} \sim N(P e_{O,t}, \Omega), \quad P = [\Sigma_e]_{MO} ([\Sigma_e]_{OO})^{-1}, \quad \Omega = [\Sigma_e]_{MM} - P [\Sigma_e]_{OM}, \quad (\text{S11})$$

and this conditional law does not depend on y_t : the state enters $e_{O,t}$ but not the regression of $e_{M,t}$ on it. This is the point at which the spatio-temporal case departs from the textbook treatment. [Durbin and Koopman \(2012\)](#) note that a missing element of the observation vector can be estimated by the corresponding element of $A_t \hat{y}_t$; that is exact when Σ_e is diagonal, in which case $P = 0$, and it is not exact here, where the exponential spatial covariance couples the locations and $P \neq 0$.

Introduce the $d \times |O|$ matrix S that lifts a vector defined on the observed rows to the full vector,

$$S_{O\cdot} = I_{|O|}, \quad S_{M\cdot} = P, \quad (\text{S12})$$

and let $\tilde{\Omega}$ be the $d \times d$ matrix with Ω in the (M, M) block and zeros elsewhere. Then (S11) can be written compactly as

$$e_t | e_{O,t} \sim N(S e_{O,t}, \tilde{\Omega}). \quad (\text{S13})$$

S2.3 The completed second moment

Write $r_{O,t} = z_{O,t} - X_{O,t}\beta - A_O \hat{y}_t$ for the residual on the observed rows, evaluated at the smoothed state. Applying the tower property with respect to $y_t | \mathcal{Z}$,

$$\mathbb{E}[e_t | \mathcal{Z}] = \mathbb{E}[\mathbb{E}[e_t | y_t, \mathcal{Z}] | \mathcal{Z}] = S r_{O,t}, \quad (\text{S14})$$

$$\text{Var}[e_t | \mathcal{Z}] = \underbrace{\mathbb{E}[\text{Var}(e_t | y_t, \mathcal{Z}) | \mathcal{Z}]}_{=\tilde{\Omega}} + \underbrace{\text{Var}[\mathbb{E}(e_t | y_t, \mathcal{Z}) | \mathcal{Z}]}_{=SA_O C_t^s A_O^\top S^\top}, \quad (\text{S15})$$

where the second term follows because, given y_t , the conditional mean of e_t is $S(z_{O,t} - X_{O,t}\beta - A_O y_t)$, an affine function of y_t with matrix $-SA_O$, whose variance under the smoothing distribution is $SA_O C_t^s A_O^\top S^\top$. Combining (S14) and (S15),

$$\boxed{\mathbb{E}[e_t e_t^\top | \mathcal{Z}] = S A_O C_t^s A_O^\top S^\top + \tilde{\Omega} + (S r_{O,t})(S r_{O,t})^\top.} \quad (\text{S16})$$

Two degenerate cases confirm the expression. If nothing is missing, then $M = \emptyset$, $S = I_d$ and $\tilde{\Omega} = 0$, and (S16) reduces to $AC_t^s A^\top + r_t r_t^\top$, the complete-data statistic. If nothing is observed, then $O = \emptyset$, S has no columns and $\tilde{\Omega} = \Sigma_e$, so (S16) returns Σ_e : with no information at that time point the measurement error keeps its unconditional variance, and contributes nothing else.

S2.4 The completed observation and the update of β

From (S14), the conditional expectation of the observation vector is

$$\hat{z}_t = E[z_t | \mathcal{Z}] = X_t \beta + A \hat{y}_t + S r_{O,t}, \quad (\text{S17})$$

which equals z_t on the observed rows, as it must, and equals $X_{M,t} \beta + A_M \hat{y}_t + P r_{O,t}$ on the missing ones. Differentiating the expected complete-data log-likelihood with respect to β and setting the derivative to zero gives

$$\hat{\beta} = \left(\sum_{t=1}^T X_t^\top \Sigma_e^{-1} X_t \right)^{-1} \sum_{t=1}^T X_t^\top \Sigma_e^{-1} (\hat{z}_t - A \hat{y}_t), \quad (\text{S18})$$

so the only change with respect to the complete-data update is the replacement of z_t by $\hat{z}_t$: the design matrix and the covariance keep their full dimension. For the same reason the divisor of the variance update remains the complete-data count dT and not the number of observed values, since (S16) already carries the conditional variance of whatever was not observed.

S2.5 Cost

The matrices P and Ω depend on the time point only through its missingness pattern, so they are computed once per distinct pattern and cached. In a monitoring network a station is typically out of service for a stretch of consecutive days, so the number of distinct patterns is far smaller than T , and the additional cost of the treatment is negligible relative to the filtering pass.

S3 Information criteria as T and d grow

This section derives what the manuscript states in words: how the likelihood-based criteria behave when the series lengthen and when the network widens, and why the two directions are not symmetric.

S3.1 Notation and the two quantities to compare

Write $q = r + 3 + 3p$ for the number of parameters of one regime, so that a K -regime configuration has $\text{df}(K) = Kq$ continuous parameters, and $n = dT$ for the number of observations entering the likelihood. Let $\hat{\ell}(K)$ be the maximized log-likelihood at K regimes, the maximum being taken over the parameters *and* over the partitions. A criterion moves from K to $K + 1$ when

$$2 \Delta \hat{\ell}(K) > \Delta \text{pen}(K), \quad \Delta \hat{\ell}(K) = \hat{\ell}(K + 1) - \hat{\ell}(K). \quad (\text{S19})$$

Everything follows from the orders of the two sides.

S3.2 The likelihood gain

Under the alternative. If the truth has $K_0 > K$ regimes, the best K -regime approximation sits at a strictly positive Kullback–Leibler distance δ_K per observation from the truth, and

$$2 \Delta \hat{\ell}(K) \asymp 2n \delta_K = O_p(dT). \quad (\text{S20})$$

The gain is proportional to the number of observations the extra regime describes: it grows in *both* d and T .

Under the null. If the truth has K regimes and we fit $K + 1$, the extra regime is superfluous. For a *fixed* refinement of the true partition the likelihood ratio is asymptotically χ_q^2 , hence $O_p(1)$ in T : lengthening the series does not help a spurious regime. What does help it is the search. The maximum is taken over the partitions of d locations into $K + 1$ groups, a set of cardinality $S(d, K + 1)$, the Stirling number of the second kind, and $\log S(d, K) = d \log K - \log K! + o(d)$. The standard extreme-value heuristic for the maximum of many χ_q^2 variables — the same heuristic that underlies the extended BIC — gives

$$2 \Delta \hat{\ell}(K) \asymp 2 [\log S(d, K + 1) - \log S(d, K)] \approx 2d \log\left(\frac{K+1}{K}\right) = O_p(d). \quad (\text{S21})$$

This is the first asymmetry, and it is the whole story in miniature: **under the null the spurious gain grows linearly in d and not at all in T** . The variables being maximized over are strongly dependent, since neighbouring partitions share most of their labels, so (S21) is an upper bound on the order rather than an equality; it is the conservative and the relevant one.

S3.3 The penalty increments

Writing $\text{df} = Kq$ and taking the increment from K to $K + 1$,

$$\Delta \text{AIC} = 2q, \quad (\text{S22})$$

$$\Delta \text{AICc} = 2q + \frac{2q^2(2K + 1)}{n} + O(n^{-2}), \quad (\text{S23})$$

$$\Delta \text{BIC} = q \log(dT) = q(\log d + \log T), \quad (\text{S24})$$

$$\Delta \text{EBIC}_\gamma = q \log(dT) + 2\gamma [\log S(d, K + 1) - \log S(d, K)] \approx q \log(dT) + 2\gamma d \log\left(\frac{K+1}{K}\right). \quad (\text{S25})$$

S3.4 Consistency, direction by direction

Combining (S21) with (S22)–(S25), a criterion avoids a spurious regime when its increment exceeds cd with $c = 2 \log\{(K + 1)/K\}$.

AIC and AICc. The increment is $2q$, a constant. The condition $2q > cd$ holds only for $d < 2q/c$, so **both are inconsistent, and the failure worsens linearly in d** . Neither depends on T at all. The corrected version adds nothing: by (S23) the correction is $O(\text{df}^2/n) = O(K^2 q^2/(dT))$ and vanishes in *both* directions. The AICc is built for the regime in which df/n is not negligible; in a spatio-temporal panel $\text{df}/n = Kq/(dT) \rightarrow 0$ however the sample grows, because df is fixed by the specification and n multiplies d by T .

BIC. The condition is $q(\log d + \log T) > cd$. Hence

- $T \rightarrow \infty$ with d fixed: the left side diverges, the right is fixed. **The BIC is consistent in T .**
- $d \rightarrow \infty$ with T fixed: the left side grows like $q \log d$, the right like cd . **The BIC over-selects, and does so for every large enough network.**
- *both growing*: consistency requires $d/\log(dT) \rightarrow 0$, that is $d = o(\log T)$ — the network must grow only logarithmically in the length of the series.

Lengthening the record therefore helps the BIC and widening the network hurts it, which is the opposite of the intuition that more data is more information.

Extended BIC. Here the penalty carries a term of the same order as the null gain. The condition becomes

$$q \log(dT) + 2\gamma d \log\left(\frac{K+1}{K}\right) > 2d \log\left(\frac{K+1}{K}\right),$$

which holds for all large d as soon as $\gamma \geq 1$. **The extended BIC is the only member of the family whose penalty has the right order in d , and $\gamma = 1$ is exactly the calibration point**, since it charges precisely the multiplicity that (S21) extracts. It follows that when the extended BIC does not change the answer on a long panel, it is not because it is miscalibrated, but because the observed gain is far above anything the null can produce.

S3.5 Criteria built on the penalized objective

Everything so far applies to criteria built on the log-likelihood. It is natural to ask what happens if they are built instead on the objective the algorithm actually maximizes,

$$Q(K, \phi) = \sum_{i=1}^d \ell_i(k_i) + \beta P(\mathcal{P}), \quad \beta = \phi c, \quad (\text{S26})$$

with $P(\mathcal{P})$ the number of neighbouring pairs that share a label. The answer depends on two scalings, and it is worth separating them.

How the Potts term scales. On an m -nearest-neighbour graph over d locations the edge set has $|E| \asymp md/2 = O(d)$ elements, so $P \leq |E| = O(d)$. When the regimes are spatially coherent, the pairs that are *not* concordant are those straddling a boundary. For K compact regions in a two-dimensional domain the total interface length is $O(L\sqrt{K})$ with L the linear size of the domain, and the number of nodes within one spacing of it is that length times the linear density $\sqrt{d}/L$. Hence

$$C(\mathcal{P}) \equiv |E| - P(\mathcal{P}) = O(\sqrt{dK}), \quad \Delta_K P = -O(\sqrt{dK}). \quad (\text{S27})$$

The concordant count is therefore of order d but its *increment* in K is only of order $\sqrt{dK}$: a boundary is a lower-dimensional object. At $\phi = 0$ the partition need not be coherent and C can be $O(d)$, which is the one regime where (S27) does not apply.

How the scale factor scales. The three options of the implementation differ exactly here. With **raw** scaling $c = 1$. With **per-observation** scaling $c = T$. With the default **auto** scaling c is the median across locations of the spread $\max_k \ell_i(k) - \min_k \ell_i(k)$, divided by the mean degree; each $\ell_i(k)$ is a sum of T terms and the per-period discrepancy between the best and the worst regime is $O(1)$, so

$$c = O(1) \text{ (raw)}, \quad c = O(T) \text{ (per-observation and auto)}. \quad (\text{S28})$$

The naive version is not a criterion. Take $\text{IC}_{\text{pen}} = -2Q + \text{pen}(\text{df}, n)$. For fixed K it is affine and decreasing in ϕ , with slope $-2cP$, so it is minimized at the largest penalty on the grid whatever the data say: it does not select ϕ , it merely records that ϕ is large. Across K the sign depends on the scaling. With c constant the increment is $2\beta \Delta_K C = O(\beta\sqrt{d}K)$, a genuine penalty on additional regimes; with the `auto` rule c is itself computed from the spread of the scores across the K candidate regimes and therefore *grows* with K , so βP increases with K and the term rewards additional regimes. Under the default scaling the comparison across K is between quantities measured on different scales, which by itself invalidates it.

The missing constant, and what the corrected version is. The deeper obstruction is that βP is the logarithm of a Potts prior on the partition only up to its normalizing constant

$$Z(\beta, K) = \sum_{\text{all } K^d \text{ labelings}} \exp\{\beta P\},$$

which depends on both quantities being selected. Restoring it changes the term to $\beta P - \log Z$. As β grows the partition function is dominated by the K constant labelings, $\log Z = \beta|E| + \log K + o(1)$, so

$$\beta P - \log Z \rightarrow -\beta C(\mathcal{P}) - \log K, \quad (\text{S29})$$

and the properly normalized criterion is

$$\text{IC}_{\text{pen}}^{\text{norm}} = \underbrace{-2\hat{\ell} + \text{pen}(\text{df}, n)}_{\text{the ordinary criterion}} + 2\beta C(\mathcal{P}) + 2\log K. \quad (\text{S30})$$

The Potts term, once normalized, is neither a reward nor a nuisance: it is a penalty proportional to the number of edges the partition cuts, that is to the length of the boundary it draws. The sign is the opposite of the naive version's, which is why the two select differently.

Orders, and what this buys. Combining (S27) and (S28), the increment of the extra term in (S30) is

$$2\beta \Delta_K C = \begin{cases} O(\sqrt{d}), & c = O(1) \quad (\text{raw}), \\ O(T\sqrt{d}), & c = O(T) \quad (\text{per-observation, auto}). \end{cases} \quad (\text{S31})$$

Set against the null gain $O(d)$ of (S21) and the alternative gain $O(dT)$ of (S20):

- with $c = O(1)$ the extra penalty is $O(\sqrt{d})$, below the null gain $O(d)$, so it does not repair the inconsistency in d and is negligible against the alternative;
- with $c = O(T)$ it is $O(T\sqrt{d})$, which dominates the null gain as soon as $T \gg \sqrt{d}$ and is smaller than the alternative gain by a factor $\sqrt{d}$. Such a criterion is therefore consistent in both directions, at the price of a detection handicap of order $\sqrt{d}$ against genuine regimes.

So a criterion on the penalized objective can be made to work, but only after restoring Z , and only in the second scaling. Two caveats remain, and they are decisive in practice. Evaluating $Z(\beta, K)$ requires path sampling at every configuration of the grid, which costs more than the grid itself. And (S30) is monotone in ϕ in the direction of coherence: raising ϕ raises the charge on every cut edge and ultimately drives the selection to $K = 1$, which has none. The criterion can rank K *given* ϕ ; it cannot choose ϕ . This is why the rule of the manuscript never builds its criterion on the penalized objective: at a given K it compares the unpenalized likelihoods that the partitions estimated at different penalties reach, and Z does not enter.

S3.6 Two effective sample sizes, and where the AICc diverges

The tension is that the model carries parameters of two kinds. The regime parameters Ψ_k are continuous and their information accrues at rate $d_k T$. The partition is discrete, has one component per location, and each label is identified by the T observations of that location: its dimension grows with d while its information per component grows with T .

If one insists on reading the partition as the object being selected, the relevant sample size is d and not dT , and the AICc correction becomes $2df(df + 1)/(d - df - 1)$. This diverges, and then changes sign, when

$$df = Kq \geq d - 1 \quad \Longleftrightarrow \quad K \geq \frac{d - 1}{q}, \quad (\text{S32})$$

which for a network of $d = 36$ locations and $q = 9$ parameters per regime, for instance, gives $K \geq 3.9$: the criterion breaks down exactly at $K = 4$. This is not a numerical accident: it is the same bound, reached from the other side, that the minimum-size constraint of the estimation algorithm imposes, since a regime needs at least $r + 2$ locations and hence $K \leq d/(r + 2)$. A network of d locations cannot identify d regime-specific parameters, whichever way one counts.

S3.7 Dependence, and the ambiguity of n

A last remark sharpens the point. All of the above uses $n = dT$, the number of *observations* (the number of observed values when the response has gaps, which is what the package uses). Under dependence that overstates the information. For a latent process with persistence G , the variance of a temporal average over T points is inflated by $(1 + G)/(1 - G)$, so an effective count is

$$T_{\text{eff}} \approx T \frac{1 - G}{1 + G}, \quad (\text{S33})$$

which at a persistence of $G = 0.97$, for instance, equals $T \times 0.015$: a series of 1,826 periods carries the information of roughly 28 independent ones. Substituting n_{eff} for n in (S24) lowers the BIC increment further, from $q \log(dT)$ to $q \log(dT_{\text{eff}})$, and makes the criterion weaker still.

The conclusion is not that one of these choices of n is correct and the others are wrong. It is that the margin of the BIC over the null gain (S21) depends on a quantity the criterion does not determine: $q \log(dT)$ with the number of observations, less with the effective count. On a persistent process and a wide network the extended BIC with $\gamma = 1$, whose multiplicity term does not depend on n , is the safer check.

S3.8 What an in-sample criterion cannot settle

There is a final, structural point, independent of the rates above. Consistency arguments require an asymptotic frame, and the two available frames answer different questions.

With $T \rightarrow \infty$ and d fixed, every label is consistently estimated, since each is identified by an unboundedly informative series, and the BIC is consistent by the argument above. But d is fixed: the model is fitted to a closed set of locations and there is no sense in which its partition extends beyond them.

With $d \rightarrow \infty$, which is the frame in which spatial prediction is meaningful, the number of labels grows with the sample. The partition is then an *incidental parameter* in the sense of [Neyman and Scott \(1948\)](#): its dimension is proportional to the sample size in the spatial direction, and it is not consistently estimated at fixed T . No penalty that is a function of df and n alone controls the error made in assigning a location to a regime, because that error does not vanish in this frame.

The choice of (K, ϕ) is made on the monitored locations, in the first frame, where the BIC is consistent. Whether the regimes extend to locations that were not monitored is a question of the second frame, which no in-sample criterion can pose; a cross-validation whose folds withhold whole locations answers it for a model already chosen, and this is why the package offers one as an optional validation (Section 4.7 of the manuscript), not as a selection rule.

S4 The ridge in the implementation: three choices, stated exactly

Section 2.7 of the manuscript writes the ridge on the original scale of the coefficients. The implementation departs from that notation in three places, each of which changes what a given value of λ means. This section states them exactly, because a reader who reproduces the estimator from (S37) below alone will not obtain the numbers the package returns.

S4.1 The scale of λ , and why the ridge needs no further one

The metric. Within a regime the coefficients enter the expected complete-data log-likelihood through the quadratic form $-\frac{1}{2}\beta' M \beta + \beta' v$, with

$$M = \sum_{t=1}^T X_t' \Sigma_e^{-1} X_t, \quad v = \sum_{t=1}^T X_t' \Sigma_e^{-1} \hat{v}_t, \quad (\text{S34})$$

where $\hat{v}_t$ is the completed observation of period t net of the smoothed latent contribution. M is a *generalized* least squares cross-product: it carries the estimated Σ_e^{-1} , so the effective design is $\Sigma_e^{-1/2} X_t$ and not X_t . Standardizing the columns of X to unit Euclidean norm, the default of every penalized-regression routine, therefore does not put the covariates on a common footing. What does is to set the diagonal of M to one. Write $s_j = M_{jj}^{1/2}$, $S = \text{diag}(s_1, \dots, s_r)$,

$$\tilde{M} = S^{-1} M S^{-1}, \quad \tilde{v} = S^{-1} v, \quad \tilde{\beta} = S \beta, \quad (\text{S35})$$

so that $\tilde{M}_{jj} = 1$. The package applies the penalty to $\tilde{\beta}$ and returns $\beta = S^{-1} \tilde{\beta}$:

$$\hat{\tilde{\beta}} = (\tilde{M} + \lambda D)^{-1} \tilde{v} \iff \hat{\beta} = (M + \lambda S D S)^{-1} v, \quad (\text{S36})$$

with D the diagonal indicator of the penalized coordinates, the intercept excluded. On the original scale the penalty is therefore $\frac{\lambda}{2} \sum_j D_{jj} M_{jj} \beta_j^2$, not $\frac{\lambda}{2} \sum_j D_{jj} \beta_j^2$ as the compact notation

$$\hat{\beta} = (M + \lambda D)^{-1} v \quad (\text{S37})$$

of the manuscript would suggest. Each coefficient is charged in proportion to the GLS information its covariate carries. The scaling is recomputed at every M-step, since Σ_e changes between EM iterations; it is undone before the coefficients are returned, so the iterates are well defined, and what varies across iterations is only the geometry in which a fixed λ acts.

What this buys. In the orthogonal case $\tilde{M} = I$ the penalized coefficient is $\tilde{\beta}_j = \tilde{v}_j / (1 + \lambda)$: the unpenalized estimate multiplied by $1/(1 + \lambda)$. Two invariances follow and are exact.

- *Units of the response.* Replacing z by az sends $v \mapsto av$ and leaves M , hence S , unchanged, so $\tilde{\beta} \mapsto a\tilde{\beta}$ and $\beta \mapsto a\beta$.

- *Level of the error variance.* Replacing Σ_e by $\sigma^2 \Sigma_e$ sends $M \mapsto M/\sigma^2$, $v \mapsto v/\sigma^2$, $s \mapsto s/\sigma$; hence $\tilde{M}$ is unchanged, $\tilde{v} \mapsto \tilde{v}/\sigma$, $\tilde{\beta} \mapsto \tilde{\beta}/\sigma$, and $\beta = S^{-1}\tilde{\beta}$ is unchanged.

So λ in (S36) is a pure number: $\lambda = 1$ halves every coefficient of an orthogonal design, whatever the units of the response and whatever the error variance of the regime.

The option `lambda_scale`, and why it does nothing to a ridge. The implementation also carries a second scaling, selected by `lambda_scale` = "relative" (the default) or "absolute", which measures a penalty against $\lambda_{\text{ref}} = \max_j |\tilde{v}_j|$, the largest partial gradient once the unpenalized coordinates have been profiled out. It exists for the L_1 part of an elastic net: a lasso penalty is a *threshold* compared with $\tilde{v}$, which carries the units of the response, and λ_{ref} removes them. It is applied to the L_1 part *only*. A ridge has no L_1 part, so for the estimator of the manuscript the two settings of `lambda_scale` return identical results. We state this explicitly because the obvious implementation, which rescales the whole penalty by λ_{ref} , is wrong for the ridge: it makes $\hat{\beta}(\lambda)$ depend on the error variance in exactly the case in which (S35) had removed that dependence, and on a design with a collinear pair of covariates the damage is of the order of 10^{-1} in the coefficients.

The conditioning ridge. Independently of λ , the implementation adds a small constant, the argument `regularization` (default 0.01), to the diagonal of the linear systems it solves, to keep them well conditioned while Σ_e is still far from its optimum. It is not part of the statistical model, and on a well-conditioned design its effect is invisible. On a collinear design it is not: it is of the order of the smallest eigenvalue of M in the offending direction, and it moves the coefficients by about 10^{-2} . Every exact comparison below is therefore run with `regularization` = 0.

S4.2 Switching off the latent process and the spatial correlation

Two switches remove parts of the model from the estimation rather than leaving the corresponding parameters to drift towards a boundary.

latent = FALSE. The loading matrix is set to $A = 0$, so the latent process no longer enters the measurement equation. Its parameters G , Σ_η and m_0 are then not identified — nothing in the data speaks about a process that loads on no location — and they are held at their input values instead of being updated by the M-step. Leaving them free is not harmless: the likelihood is flat in them, and the EM algorithm spends its iterations wandering on that flat ridge.

spatial = FALSE. The exponential correlation $\exp(-\theta h)$ is replaced by the identity, so $\Sigma_e = (\sigma_\epsilon^2 + \sigma_\omega^2)I$. Only the sum of the two variance components is then identified, and the range θ has no meaning; the Newton–Raphson update of $(\theta, \sigma_\epsilon^2/\sigma_\omega^2)$ is skipped rather than run on a direction the likelihood does not constrain. The alternative, letting $\sigma_\omega^2 \rightarrow 0$, is not available by construction: the internal parameterization is $\log(\sigma_\epsilon^2/\sigma_\omega^2)$, so a zero partial sill is a boundary the algorithm can approach but never reach, and in our experiments it chased that boundary for minutes on problems that otherwise converge in a second.

The exact special case. With both switches off and `regularization` = 0 the model is the Gaussian linear regression $z_{it} = x'_{it}\beta + e_{it}$ with independent errors, $M = \sigma^{-2} \sum_t X'_t X_t$, and (S36) is the ridge estimator on columns standardized to unit $X'X$ diagonal. The package reproduces it to machine precision: the ordinary least squares solution at $\lambda = 0$ to 10^{-11} , and the ridge at $\lambda = 0.3$ to 4×10^{-14} . With $K > 1$ the same switches turn SC-STEM into a spatially clustered

linear regression with a Potts penalty on the labels, which is the natural nested reference against which the contribution of the latent process and of the spatial covariance to an application can be measured: fit the model with and without each component and compare them out of sample.

S4.3 One λ for all regimes, or one strength per regime

With k regimes the penalty is applied within each, $\hat{\beta}_h = (\tilde{M}_h + \lambda_h D)^{-1} \tilde{v}_h$, and the question is how λ_h relates to the single hyperparameter λ . The implementation offers two answers through the argument `lambda.by`; both keep *one* hyperparameter, and they differ in what that hyperparameter holds fixed across regimes.

"common" (the default): the same proportional shrinkage. $\lambda_h = \lambda$ for every h . Because the metric (S35) is computed within each regime, $\tilde{M}_h$ has unit diagonal whatever the size of the regime, so in the orthogonal case every regime's coefficients are multiplied by the same factor $1/(1 + \lambda)$. On the original scale the penalty on coefficient j of regime h is $\frac{\lambda}{2} M_{h,jj} \beta_{hj}^2$, and since $M_{h,jj} \approx n_h T \bar{x}_j^2 / \sigma_h^2$ grows with the number of locations n_h of the regime, the implied Gaussian prior on β_{hj} has precision proportional to the information the regime carries — a *unit-information* prior, in the sense of the g -prior of Zellner (1986) with g proportional to the sample size.

"size": the same prior. $\lambda_h = \lambda \bar{n} / n_h$, with $\bar{n} = d/k$ the average regime size. The prior precision on the original scale becomes $\lambda \bar{n} M_{h,jj} / n_h \approx \lambda \bar{n} T \bar{x}_j^2 / \sigma_h^2$, which no longer depends on n_h : every regime is shrunk towards zero by the *same* prior, and a regime of half the average size is shrunk proportionally twice as hard, because its data carry half the information. A penalty of this kind is sometimes described as a penalty per observation.

Which one tracks the optimum. In the orthogonal case with unit error variance on the standardized scale, the ridge parameter that minimizes the mean squared error of $\tilde{\beta}_{hj}$ is $\lambda_{hj}^* = 1/\tilde{\beta}_{hj}^2 = 1/(M_{h,jj} \beta_{hj}^2)$. With comparable coefficients across regimes this is proportional to $1/M_{h,jj}$, that is to $1/n_h$ at fixed T :

$$\lambda_h^* \propto \frac{1}{n_h}. \quad (\text{S38})$$

The **"size"** rule reproduces exactly this dependence with a single hyperparameter; the **"common"** rule ignores it, and therefore under-shrinks the small regimes and over-shrinks the large ones. The difference is immaterial with balanced regimes, where $n_h = \bar{n}$ and the two coincide, and it grows with the imbalance. Neither makes the penalty genuinely regime-specific, which would require one hyperparameter per regime, a grid of size $|A|^k$, and a selection rule that a regime of a dozen locations cannot support; that construction is developed elsewhere and is not used in the manuscript.

S5 Which step comes first: the ridge, the penalty and the number of regimes

With the ridge in force the procedure carries three quantities that are chosen rather than estimated: the number of regimes K , the strength ϕ of the spatial penalty on the partition, and the strength λ of the ridge on the coefficients. A practitioner's first question is whether they can be chosen in sequence — fix λ on the pooled fit, then look for (K, ϕ) , or the reverse. They cannot, and this section says exactly why and in which direction each coupling acts.

S5.1 The objective, and where each hyperparameter enters

With $g = (g_1, \dots, g_d)$ the labels and E the edges of the neighbourhood graph, the objective is

$$\mathcal{Q}(g, \Psi; K, \phi, \lambda) = \underbrace{\sum_{i=1}^d \ell_i(g_i; \Psi_{g_i})}_{\text{fit}} + \underbrace{\phi c \sum_{(i,j) \in E} \mathbb{1}\{g_i = g_j\}}_{\text{Potts}} - \underbrace{\frac{1}{2} \sum_{h=1}^K \lambda_h \tilde{\beta}'_h D \tilde{\beta}_h}_{\text{ridge}}. \quad (\text{S39})$$

ϕ and λ are weights inside (S39) and change the estimate; K is the dimension of the model. The two penalties act on different arguments — the Potts term on the labels, the ridge on the parameters — and that asymmetry is the whole content of what follows.

Proposition S1 (the ridge leaves the label step unchanged). *At fixed Ψ the map $g \mapsto \mathcal{Q}(g, \Psi)$ differs from its unpenalized counterpart by a constant, so the label update*

$$g_i \leftarrow \arg \max_h \left\{ \ell_i(h; \Psi_h) + \phi c \sum_{j: (i,j) \in E} \mathbb{1}\{h = g_j\} \right\}$$

is the same with and without the ridge.

Proof. Moving location i from regime h to regime h' changes neither β_h nor $\beta_{h'}$, which the label step holds fixed, so the third term of (S39) does not depend on g and drops out of every arg-max. (With `lambda_by = "size"` the weights λ_h depend on the sizes n_h and hence on g ; they are, however, evaluated at the sizes of the parameter step and held fixed through the label step, exactly as the parameters are.) $\square$

The ridge therefore reaches the partition only *through* the parameters the labels are scored against. That is not a negligible channel: shrinkage moves every β_h towards the same point, the spread of $\ell_i(\cdot)$ across regimes narrows, and with it the evidence any single location carries about its own label.

S5.2 Three couplings, in the direction they act

λ depends on K . By (S38) the optimal strength in a regime of n_h locations is proportional to $1/n_h$; with K balanced regimes $n_h = d/K$, so

$$\lambda^* \propto \frac{K}{d}. \quad (\text{S40})$$

The optimal shrinkage *grows* with K , linearly. A λ calibrated on the pooled network is too small by a factor of about K for the regimes it will be used in: it under-shrinks exactly where collinearity bites hardest.

K depends on λ . Criteria count the effective number of coefficients $\text{df}_h(\lambda) = \text{tr}\{\tilde{M}_h(\tilde{M}_h + \lambda_h D)^{-1}\}$, which decreases from r at $\lambda = 0$ to the number of unpenalized coordinates as $\lambda \rightarrow \infty$. The price of an additional regime under the BIC,

$$\Delta_K = \log(dT) \{ \text{df}(\lambda) + q \}, \quad (\text{S41})$$

with q the number of parameters of a regime other than the coefficients, is therefore *decreasing* in λ : shrinkage makes an extra regime cheaper and pushes the selected K up. The two dependencies run in opposite directions, so neither can be resolved without the other.

ϕ depends on λ , through its scale. The default scale of the Potts term is $c = \text{med}_i \{ \max_h \ell_i(h) - \min_h \ell_i(h) \} / \bar{m}$, the typical gain from picking the best-fitting regime, per neighbour (Section 4.5 of the manuscript). Since shrinkage narrows the spread of the scores, c is decreasing in λ : a *fixed* ϕ buys less spatial smoothing at a larger λ . Two points of a grid indexed by (ϕ, λ) are comparable in ϕ only if c is held fixed across them, which is what `phi_scale = "raw"` does; under the default each pair (K, λ) calibrates its own c on its unpenalized fit (Section S6), and a joint grid must be read with that in mind.

S5.3 What follows for the search

Joint. Traverse the whole grid (K, ϕ, λ) and select on it. Correct; the cost is the product of the three grid sizes.

Profiled, which we recommend. For each (K, ϕ) select λ ; then compare the (K, ϕ) configurations, each at its own $\hat{\lambda}(K, \phi)$. The cost is that of the joint search, but the rule is applied hierarchically, which matters because ϕ and λ enter the criterion differently: the Potts term never enters it, since its normalizing constant $Z(\phi, K)$ is intractable (Section S3.5), and ϕ is compared only through the unpenalized likelihoods its partitions reach at a given K , while λ enters through the effective degrees of freedom. The λ surface is flat relative to the K surface, so a coarse logarithmic grid suffices.

Sequential, and why not. Fixing λ on the pooled fit and only then searching (K, ϕ) is cheap and wrong in the specific direction of (S40): it applies the shrinkage appropriate to d locations inside regimes of d/K . The reverse order, fixing (K, ϕ) unpenalized and then choosing λ , is wrong in the direction of (S41): the unpenalized fit charges every regime its full r coefficients and so selects too few regimes for the λ eventually used.

S5.4 Which likelihood the criterion sees

It is tempting to select on the objective (S39) *including* the ridge term. That double-counts: the reduction in flexibility the ridge produces is already carried by $\text{df}(\lambda)$, and subtracting $\frac{1}{2} \lambda \tilde{\beta}' D \tilde{\beta}$ from the log-likelihood as well charges for it twice, once through the objective and once through the dimension. The construction consistent with the effective-degrees-of-freedom literature is

$$\text{IC} = -2 \hat{\ell}(\hat{\Psi}(\lambda)) + \text{pen}(dT) \left\{ \sum_{h=1}^k \text{df}_h(\lambda) + kq \right\}, \quad (\text{S42})$$

with $\hat{\ell}$ the *unpenalized* log-likelihood evaluated at the *penalized* estimate. The coherent alternative is Bayesian: read the ridge as a Gaussian prior and select on the marginal likelihood, which for a Gaussian prior is available in closed form. The Potts term admits the same reading only up to $Z(\phi, K)$, which is not available, so a procedure that marginalized over β while conditioning on ϕ would apply two different logics to two penalties of the same objective. We do not pursue it.

S6 Where a fit starts, and the scale of its penalty

Section 4.1 of the manuscript states two rules of initialization. An unpenalized fit starts from the k -means of the departures of the locations from the pooled model; a fit with $\phi > 0$ starts from the partition to which the fit with $\phi = 0$ at the same K has converged. The alternating algorithm is otherwise unchanged. This section defines the departures and says why they, and not the covariate means, carry information on the regimes; then it states what the second rule does to the scale of

the penalty, why a penalized fit cannot be started like an unpenalized one, and what follows for the grid of ϕ .

S6.1 The start of an unpenalized fit: departures from the pooled model

Let $\hat{\mu}_{it} = x'_{it}\hat{\beta} + K_i\hat{y}_t$ be the signal of the pooled fit and $r_{it} = z_{it} - \hat{\mu}_{it}$ the residual of location i . With $\tilde{x}_{it}$ the covariates that vary, regress r_{it} on $(1, \tilde{x}_{it})$ over the periods at which i is observed, with slopes b_i and residuals u_{it} . The features of location i are

$$f_i = \left(\bar{r}_i, b_i, \text{corr}(u_{it}, u_{i,t-1}), \log \widehat{\text{Var}}(u_{it}) \right), \quad (\text{S43})$$

with $\bar{r}_i$ the time mean of r_{it} . The partition is the k -means of the f_i , standardized and compressed by principal components, with restarts and the admissibility filter of Section 4.1.

Why departures and not covariate means. Under the SC-STEM model, for a location of regime g ,

$$r_{it} = x'_{it}(\beta_g - \hat{\beta}) + K_i(y_t^{(g)} - \hat{y}_t) + e_{it},$$

so the mean and the slopes in (S43) estimate how β_g departs from the pooled coefficients, the autocorrelation of u_{it} reflects the dynamics of $y_t^{(g)} - \hat{y}_t$, and its variance the amplitude of that path together with the error variance of the regime. Every term is common to the locations of one regime and differs between regimes whenever their parameters, or their latent paths, differ. The covariate mean $\bar{x}_i$, which the earlier default clustered, depends on the regime only if the covariates do. When the covariates are exogenous to the regimes — the same distribution in every regime, which is what a regression on them usually presumes — $\bar{x}_i$ has the same law in every regime, and its k -means returns a partition unrelated to the regimes, from which the alternation has to find its way on its own.

Cost. On a grid of (K, ϕ) the pooled fit is the member at $K = 1$, and (S43) is computed once from it; a single fit at $K > 1$ fits the pooled model first.

S6.2 The rule

Fix K and write $\mathcal{A}(g^{(0)}, \phi)$ for Algorithm 1 of the manuscript started from the partition $g^{(0)}$ at penalty ϕ . Let g^{init} be the partition given by the initialization of Section 4.1 (the k -means on the covariate means, or a partition supplied by the user). The estimates are

$$(\hat{g}^0, \hat{\Psi}^0) = \mathcal{A}(g^{\text{init}}, 0), \quad (\hat{g}^\phi, \hat{\Psi}^\phi) = \mathcal{A}(\hat{g}^0, \phi) \quad (\phi > 0).$$

In the package, `SCSTEM_Estimation()` with $\phi > 0$ and no starting partition computes $\hat{g}^0$ first, with the same arguments; `SCSTEM_Infocrit()`, which visits $\phi = 0$ first at every K , passes $\hat{g}^0$ to the penalized fits as their starting partition.

S6.3 The scale is the one of the unpenalized fit

The scale of the penalty under the default normalization is computed at the first sweep,

$$c = \frac{1}{\bar{m}} \text{med}_i \Delta_i, \quad \Delta_i = \max_h \ell_i(h; \Psi_h^{(1)}) - \min_h \ell_i(h; \Psi_h^{(1)}), \quad (\text{S44})$$

with $\Psi^{(1)}$ the estimates of the first parameter step and $\bar{m}$ the average number of neighbours.

Proposition S2. *Under the rule, $\Psi^{(1)} = \hat{\Psi}^0$, so c in (S44) is computed on the fitted unpenalized regimes. Consequently: (i) c is the same for every $\phi > 0$ at the same K ; (ii) c and the penalized fits depend on the initialization only through $\hat{g}^0$, so two initializations whose unpenalized fits converge to the same partition give identical penalized fits; (iii) a single penalized fit coincides with the member of a grid at the same (K, ϕ) .*

Proof. The first parameter step of $\mathcal{A}(\hat{g}^0, \phi)$ estimates each regime by the EM algorithm on the locations of $\hat{g}^0$ in that regime, from the same starting values and with the same tolerance as the final refit of $\mathcal{A}(g^{\text{init}}, 0)$, which is the same computation on the same partition. The EM algorithm is a deterministic map of the data, the partition and the starting values, so $\Psi^{(1)} = \hat{\Psi}^0$, and (S44) depends on $(\hat{g}^0, \hat{\Psi}^0)$ only. Neither depends on ϕ , which gives (i); both depend on g^{init} only through $\hat{g}^0$, which gives (ii); and a grid passes exactly $\hat{g}^0$, which gives (iii). $\square$

What the scale means. At fixed parameters, location i leaves its best unpenalized regime g^* for a regime h only if

$$\ell_i(g^*) - \ell_i(h) < \phi c \{n_i(h) - n_i(g^*)\} = \phi \text{med}_j \Delta_j \frac{n_i(h) - n_i(g^*)}{\bar{m}}, \quad (\text{S45})$$

with $n_i(h)$ the number of neighbours of i in regime h . With all its neighbours in h the right-hand side is about $\phi \text{med}_j \Delta_j$: the location moves only if its own advantage for g^* is below the fraction ϕ of the typical best-against-worst advantage. On fitted regimes Δ_j is of the order of the whole log-likelihood advantage of the right regime over the wrong ones accumulated over the T periods, so values of ϕ of a few hundredths already move the locations whose assignment is ambiguous, and values of order one override the likelihood for most of those near a border. This is why the default grid is $\phi \in \{0, 0.025, 0.05, 0.1, 0.2, 0.5, 1\}$ and the moderate-penalty band in which the selection rule chooses K is $[0.025, 0.2]$.

S6.4 Why a penalized fit cannot be started like an unpenalized one

Two things go wrong when the penalized fit is started from g^{init} instead.

The scale depends on the start. If g^{init} is unrelated to the regimes, every starting regime is a mixture of the true ones, the estimates $\Psi_h^{(1)}$ are nearly alike, and the spreads Δ_i in (S44) are small; if g^{init} is close to the regimes, the spreads are those of distinct regimes and several times larger. The same ϕ then stands for penalties of different strength under different initializations, and a better initialization receives a stronger penalty. This was the behaviour of the package before the rule, and it made the comparison of initializations at a fixed ϕ meaningless.

A strong penalty freezes a poor start. Suppose instead that c is fixed on fitted regimes but the fit starts from a poor g^{init} . At the first sweep the scores of a location barely differ across the nearly alike starting regimes, while the penalty term is of the order of $\phi \text{med}_j \Delta_j$. When

$$\phi c > \max_i \max_{h, h'} |\ell_i(h) - \ell_i(h')|,$$

every location whose neighbours are not tied between two labels takes the label held by most of its neighbours, whatever its scores: the label step reduces to the iterated conditional modes of the Potts model alone, a majority vote whose fixed points are the configurations in which every location sits in the regime of most of its neighbours. The start is smoothed into compact blocks, the parameters re-estimated on those blocks are again mixtures, and the alternation stops at a fixed point near the

start. From $\hat{g}^0$ the regimes are already as distinct as the data make them, the scores separate the locations, and the penalty only arbitrates the ambiguous ones through (S45).

S6.5 Cost

On a grid of (K, ϕ) that contains $\phi = 0$ the rule costs nothing: the unpenalized fit at every K is a member of the grid and its partition is passed on. A single penalized fit costs one unpenalized fit more. In the parametric bootstrap every refit at $\hat{\phi} > 0$ fits $\phi = 0$ first on the resampled data, which doubles its cost; that is the estimator reproduced, since the start and the scale are part of it.

S7 Simulation experiments

All the material of the simulation study — the data-generating process, the scenarios and their parameter values, the separation of the regimes, the design, the computational settings, the time limit of a replication, the cost and the extended results — is collected in Supplementary Material B.

S8 Reproduction code

Everything in the paper is produced by the **Stem** package (Cameletti et al., 2026) and by a set of standalone scripts, the replication material, which is distributed separately from the package and needs nothing but an installed **Stem**. Each script runs from whatever folder it sits in, writes its results beside itself, checks that the installed **Stem** carries the functions it uses, and installs or updates it otherwise.

`run-simulations.R` runs the simulation study of Section 3 of the manuscript.

`design-figures.R` produces the figures and tables that describe the design of the study, in Supplementary Material B.

`analyse-simulations.R` turns the results into the tables and figures of the manuscript and of Supplementary Material B.

Every routine that uses randomness takes an explicit seed and restores the state of the random number generator on exit, so the experiments reproduce exactly. The package and the replication material are available at <https://github.com/PaoloMaranzano/Stem>.

References

- Michela Cameletti, Francesco Caccia, and Paolo Maranzano. *Stem: Spatio-Temporal Expectation-Maximization Models and their Spatially-Clustered Extension*, 2026. URL <https://github.com/PaoloMaranzano/Stem>. R package version 2.0.0.
- Piet de Jong. The diffuse Kalman filter. *The Annals of Statistics*, 19(2):1073–1083, 1991. doi: 10.1214/aos/1176348139.
- Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1):1–22, 1977.
- Yu Du and Ravi Varadhan. SQUAREM: An R package for off-the-shelf acceleration of EM, MM and other EM-like monotone algorithms. *Journal of Statistical Software*, 92(7):1–41, 2020. doi: 10.18637/jss.v092.i07.

- James Durbin and Siem Jan Koopman. *Time Series Analysis by State Space Methods*. Oxford University Press, Oxford, 2nd edition, 2012.
- Chuanhai Liu and Donald B. Rubin. The ECME algorithm: a simple extension of EM and ECM with faster monotone convergence. *Biometrika*, 81(4):633–648, 1994. doi: 10.1093/biomet/81.4.633.
- Xiao-Li Meng and Donald B. Rubin. Maximum likelihood estimation via the ECM algorithm: a general framework. *Biometrika*, 80(2):267–278, 1993. doi: 10.1093/biomet/80.2.267.
- Xiao-Li Meng and Donald B. Rubin. On the global and componentwise rates of convergence of the EM algorithm. *Linear Algebra and its Applications*, 199:413–425, 1994. doi: 10.1016/0024-3795(94)90363-8.
- Xiao-Li Meng and David van Dyk. The EM algorithm – an old folk-song sung to a fast new tune. *Journal of the Royal Statistical Society, Series B*, 59(3):511–567, 1997. doi: 10.1111/1467-9868.00082.
- Jerzy Neyman and Elizabeth L. Scott. Consistent estimates based on partially consistent observations. *Econometrica*, 16(1):1–32, 1948. doi: 10.2307/1914288.
- Christophe Roland and Ravi Varadhan. New iterative schemes for nonlinear fixed point problems, with applications to problems with bifurcations and incomplete-data problems. *Applied Numerical Mathematics*, 55(2):215–226, 2005. doi: 10.1016/j.apnum.2004.10.003.
- Robert H. Shumway and David S. Stoffer. An approach to time series smoothing and forecasting using the EM algorithm. *Journal of Time Series Analysis*, 3(4):253–264, 1982. doi: 10.1111/j.1467-9892.1982.tb00349.x.
- Ravi Varadhan and Christophe Roland. Simple and globally convergent methods for accelerating the convergence of any EM algorithm. *Scandinavian Journal of Statistics*, 35(2):335–353, 2008. doi: 10.1111/j.1467-9469.2007.00585.x.
- Yaqiong Wang, Francesco Finazzi, and Alessandro Fassò. D-STEM v2: A software for modeling functional spatio-temporal data. *Journal of Statistical Software*, 99(10):1–29, 2021. doi: 10.18637/jss.v099.i10.
- Arnold Zellner. On assessing prior distributions and Bayesian regression analysis with g -prior distributions. In Prem K. Goel and Arnold Zellner, editors, *Bayesian Inference and Decision Techniques: Essays in Honor of Bruno de Finetti*, pages 233–243. North-Holland, Amsterdam, 1986.